

Activity Guide — middle / module-10-herbs-aushadhi

IKS Curriculum

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Activity 01 — Smell-Identification Challenge

Module: Auṣadhi · **Band:** Middle · **Day:** woven through Days 2–7 (mainly Day 7 for the assessment version) · **Time:** 15 min total

Type

Sensory identification, individual then paired

Goal

Each student can identify all six module herbs by smell alone, eyes closed.

Materials per group

- 6 small opaque cups (paper cups work) per group of 4 students — one cup per herb
- Crushed/torn samples of each herb pressed into the bottom of each cup:
 - tulsi leaf
 - dried neem leaf (sealed cup — smell through small hole only)
 - turmeric powder
 - ginger powder OR a slice of fresh ginger
 - ajwain seeds (whole, lightly crushed)
 - mint leaf
- 1 numbered worksheet per student

Set-up (teacher, before class)

- Label each cup with a number (1–6). Same numbering across groups, randomised so students can't compare with their neighbour.
- Keep a master key for grading.
- Cover cups with a piece of paper held by a rubber band, with a small hole pierced for smelling.

Instructions (student-facing)

Smell-Identification Challenge — individual round, then pair discussion.

1. Take the 6 cups assigned to your table.
2. Smell each cup through the hole in the cover. Do **not** open the cups. Do **not** taste anything.
3. On your worksheet, write the Sanskrit name and one feature you noticed for each numbered cup.
4. After 5 minutes — turn to your pair partner and compare answers. **Don't change yours.** Just compare.
5. Then the teacher reveals the answers. Self-score on the back of the worksheet.

Worksheet

Cup #	Sanskrit name	One feature
1		
2		
3		
4		
5		
6		

Self-score (after reveal): ___ / 6

Discussion (last 3 min)

- Which herb was hardest to identify? Why?
- Was there a herb you smelled and immediately knew? Why was it easy?
- Did anyone confuse two herbs? Which pair? What do they have in common?

(Common confusion: tulsi and mint, both square-stem Lamiaceae; tulsi and clove, both eugenol-dominated; ajwain and dried thyme, both thymol-dominated.)

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Students smell each cup, look thoughtful, write a few words per herb	“This smells like clove — must be tulsi” / “Sharp! Like medicine. Ajwain?”
Needs intervention	Students rushing through; just writing “I don’t know”	Silence; or “I’ll just copy you”

Safety / sensitivity

- **No opening cups.** No tasting.
- **Neem cup smell only.** No touching the inside.
- **Allergy check before activity** — students with asthma or known plant allergies can opt out of smelling specific cups.
- **Don't put cups directly under anyone's nose** — let students choose their own distance.

Differentiation

- **Tier up:** A pair gets 8 cups (the 6 plus 2 confusers — sweet basil, dried thyme). They must identify which are the 6 module herbs and which are imposters.
- **Tier down:** Only 4 cups (tulsi, ginger powder, mint, ajwain) — the most distinctive smells.

Clean-up

- Seal all cups for next class (this activity can be reused for revision).
- Teacher checks if any cup has degraded (mint dries out fast — refresh as needed).
- Worksheets stored in the project folder.

Activity 1 — Blind Smell-Test Identification

Module: Herbs — Auṣadhi · **Band:** Middle · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Herbs — Auṣadhi through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

small samples of each of the six herbs, blindfolds (optional), notebook

What students do

Students close their eyes and try to identify each herb by smell. They record their guess + observation, then check.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.

- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 02 — Sort by *Vīrya*

Module: Auśadhi · **Band:** Middle · **Day:** 8 · **Time:** 15 min (within Day 8’s 45 min)

Type

Card-sort, paired, with defended reasoning

Goal

Each pair correctly sorts 12 kitchen items into heating (*uṣṇa*) / cooling (*śīta*) / not-sure, and articulates the **reasoning** behind their classification — using both sensory observation and the Ayurvedic logic of taste-and-effect.

Materials per pair

- 12 cards, each with one item name (no answer printed):
 - **6 module herbs:** tulasī, nimba, haridrā, adraka, yavānī, pudīnā
 - **6 kitchen-pantry items:** rice, ghee, lemon, ice cream, red chilli, coriander seed
- 1 sorting mat with three columns labelled *Uṣṇa* (heating) / *Śīta* (cooling) / *Not sure*
- 1 worksheet per pair (below)

Set-up (teacher, before class)

- Pre-shuffle each pair’s 12 cards.
- Have the sorting mats ready on each desk.

Instructions (student-facing)

Sort by *Vīrya* — work in pairs.

1. Take your 12 cards and your sorting mat.
2. For each item, decide together: heating (*uṣṇa*), cooling (*śīta*), or not sure.
3. Fill in the worksheet — name each item, your choice, and **one sentence explaining why**.
4. **Disagree with your partner?** Argue it out before you fill in the row. Use sensory evidence (“ginger feels warm when I eat it”) or the doṣa logic (“bitterness reduces pitta, so neem is functionally cooling”).
5. Last 3 min: gallery walk — visit two other pairs’ mats. Note one card they placed differently from you.

Worksheet (one per pair)

#	Item	Heating / Cooling / Not sure	Reasoning (one sentence)
1	tulasī		
2	nimba		
3	haridrā		
4	adraka		
5	yavānī		
6	pudīnā		
7	rice		
8	ghee		
9	lemon		
10	ice cream		
11	red chilli		
12	coriander seed		

Teacher key (do not show to students until after sort)

Item	Classification	Note
tulasī	<i>uṣṇa</i> (heating)	Eugenol- dominated aromatic warming
nimba	<i>śīta</i> (cooling)	Functional: bitterness reduces pitta
haridrā	mild <i>uṣṇa</i> (heating)	Curcumin; some texts call it neutral
adraka	<i>uṣṇa</i> (heating)	Gingerol activates TRPV1 (warm)
yavānī	<i>uṣṇa</i> (heating)	Thymol pungent stimulant
pudīnā	<i>śīta</i> (cooling)	Menthol activates TRPM8 (cool)
rice	<i>śīta</i> (cooling)	White rice in classical texts; basmati specifically <i>śīta</i>

Item	Classification	Note
ghee	<i>śīta</i> (cooling)	Classical Ayurvedic classification — interesting because it’s warming sensation but cooling potency
lemon	<i>uṣṇa</i> (heating)	Sour and heating in classical texts (some debate; some texts call it neutral)
ice cream	<i>śīta</i> (cooling)	Both sensory and functional
red chilli	<i>uṣṇa</i> (heating)	Capsaicin activates TRPV1
coriander seed	<i>śīta</i> (cooling)	One of the few <i>śīta</i> spices — used to balance heat

Note for the teacher: the “interesting” ones are **ghee** (warming feeling but cooling potency in classical texts) and **lemon** (sour and somewhat heating despite cooling lemonade culture). These are where students will argue most productively. The argument *is* the learning.

Discussion prompts (last 3 min, whole class)

- Which item caused the most disagreement in your pair?
- Did anyone classify ghee as cooling? Why might that be?
- What’s the difference between “feels warm/cool” and “has a warming/cooling effect over time”? (This is the sensory vs functional point.)

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Two partners pointing at cards, both holding the sorting mat, sometimes laughing about ghee	“But ghee feels warm!” / “Yes but it cools pitta...” / “Maybe both? Let’s go with cooling but write ‘tricky’”
Needs intervention	One partner sorting silently; cards being placed without discussion	Silence or “I don’t care, put it anywhere”

Safety / sensitivity

- No food handling in this activity (it's card-only).
- If a student raises dietary or religious restrictions about any item (e.g. ghee, ice cream), acknowledge and continue — the sort is a classification exercise, not a food choice.

Differentiation

- **Tier up:** Add 4 more cards — *aśvagandhā* (heating; reduces vāta), *brāhmī* (cooling; pitta-reducing), green tea (cooling), black pepper (heating). Pair must research one of the unfamiliar items before classifying.
- **Tier down:** Use only 8 cards (the 6 herbs + ice cream + chilli) — the most clearly classifiable.

Clean-up

- Cards collected and stored in card bag for future use.
- Worksheets stapled into the student workbook.

Activity 03 — Herb Deck Build

Module: Auśadhi · **Band:** Middle · **Day:** 9 (and finalised Day 10 morning) · **Time:** 35 min in-class + variable at home

Type

Project artefact construction, paired

Goal

Each pair produces six A6 (or quarter-A4) cards — one per module herb — with all eight required elements per card. The deck becomes the pair's project submission and a classroom resource after Day 10.

Materials per pair

- 8 blank cards (6 + 2 spares)
- Colour pencils, fine-tip markers, ruler
- Reference materials (their own Days 2–8 worksheets)
- 1 envelope or rubber band for storing the finished deck

The card template (eight elements per card)

+-----+	
	[SKETCH OF HERB]
	Sanskrit (IAST): _____
	English / Common: _____
	Botanical name: _____

Family:	_____
Identifying features:	
1.	_____
2.	_____
3.	_____
Active compound:	_____
Traditional use:	_____
Safety note:	_____

Instructions (student-facing)

Today: build the deck.

1. Lay out your six blank cards. Decide who draws which (3 each, then swap and review).
2. Fill in the eight elements per card. Use your worksheets — don't re-invent.
3. **Sketch first** (least time-consuming), **safety note last** (most time-consuming to get right).
4. Check each card against the rubric before finalising. The rubric is on the wall.
5. By the end of class, all six cards should be drafted. Refine at home tonight.

Quality checklist for each card (rubric-aligned)

- ☐ Sketch is clear enough that someone who's never seen the herb could recognise it
- ☐ Sanskrit name in IAST (with diacritics)
- ☐ Botanical name italicised, correctly spelled
- ☐ Family named (one of: Lamiaceae, Apiaceae, Zingiberaceae, Meliaceae)
- ☐ Three distinct identifying features (not three versions of the same feature)
- ☐ Active compound correctly named (eugenol, azadirachtin, curcumin, gingerol, thymol, menthol)
- ☐ One specific traditional use (no vague claims)
- ☐ One specific safety note (no vague reassurances)

Common failure modes (teacher watches for these)

1. **Sketching one card for 20 minutes.** Cap at 5 minutes per sketch.
2. **Vague traditional use** ("good for health"). Demand specificity: "*good for what — relieving congestion? helping digestion? cooling in summer?*"
3. **Vague safety note** ("be careful"). Demand specificity: "*be careful of what — concentration? dose? interactions?*"
4. **Two partners not coordinating.** If you see this — have them switch sides at minute 15.

Differentiation

- **Tier up:** A 7th bonus card (any herb), applying the same eight-element template — *aśvagandhā*, *brāhmī*, *triphalā*, *kariveppilai*, *dhānyaka*. Plus: write a 30-second pitch for the bonus card on the back.

- **Tier down:** Reduce to 4 cards (tulsi, ginger, turmeric, mint) but each must have all 8 elements complete. Quality over quantity.

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners drawing/writing; checking the rubric on the wall; quick discussion before each entry	“What’s the safety note for ginger?” / “Heartburn at high doses?” / “Yes — write that.”
Needs intervention	One partner doing everything; cards being copied verbatim from worksheets without engagement	Silence; or “let’s just write ‘good for cold’”

Clean-up

- Each pair stores their cards in an envelope with both names + class on the front.
- Refine at home overnight; bring on Day 10.

Project Brief

Project Brief — Class Herb Identification Deck

Module: Herbs — Auśadhi · **Band:** Middle

What you will do

Each student creates one identification card for one of the six herbs: Sanskrit/botanical name, leaf drawing, smell description, three traditional uses, one active compound, one safety note. Cards combined into class library.

Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn’t.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (300 words (Middle)): what you learned, what surprised you, what you’re still uncertain about.
- **Presentation** (5-minute share with the class).

Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

Sources

You must cite at least 2 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.